DATASET INTELLIGENCE & OPERATIONAL ANALYSIS

Domain: Universal Dataset Intelligence & Distribution Analysis (0.0%) | Profiling Conf: 91.7% | Records: 500 | Hash: bab4e166

Filter: ALL_RECORDS(UNFILTERED) | Audit: VERIFIED_DETERMINISTIC_PARITY | Engine: v8-javascript

KEY PERFORMANCE INDICATORS (DETERMINISTIC METRIC AUDIT)

TOTAL RECORDS VOLUME

500

`COUNTROWS`

Omitted: 0 rows

UNIQUE ENTITIES (EMP_ID)

500

`COUNTDISTINCT(Emp_ID)`

Omitted: 0 rows

TOTAL BONUS_AMOUNT

5,059,110

`SUM(Bonus_Amount)`

Omitted: 0 rows

MEAN BONUS_AMOUNT

10,118.22

`AVERAGE(Bonus_Amount)`

Omitted: 0 rows

MEDIAN BONUS_AMOUNT

10,118

`MEDIAN(Bonus_Amount)`

Omitted: 0 rows

TOTAL ANNUAL_SALARY

42,159,250

`SUM(Annual_Salary)`

Omitted: 0 rows

MEAN ANNUAL_SALARY

84,318.50

`AVERAGE(Annual_Salary)`

Omitted: 0 rows

MEDIAN ANNUAL_SALARY

84,318.50

`MEDIAN(Annual_Salary)`

Omitted: 0 rows

EXECUTIVE MANAGEMENT DECISION BRIEF

- **Ingestion & Data Quality Profile:** Audited 500 source rows across 9 columns with 100.0% full record completeness. Detected 0 duplicate records in source data.
- **Total Records Volume Benchmark:** Total Records Volume stands at 500 calculated via COUNTROWS. Supporting benchmark Mean Bonus_Amount stands at 10,118.22.
- **Cohort Concentration (Department):** Largest volume resides in 'Sales' with 8,50,621 (16.8% of cohort total). Lead cohort exceeds 'Human Resources' by 1.0x.
- **Temporal Trajectory (Month):** Measured from 2021-01-01 (4,15,858) to 2024-12-01 (4,15,184) with -0.2% net trajectory. Peak volume recorded in 2021-09-01 (4,27,382).

DATASET INTEGRITY & STATISTICAL COVERAGE

Audited Records: 500 | Fully Populated: 500 (100.0%) | Duplicates: 0 | Missingness Exclusions Documented

VISUAL COHORT BREAKDOWNS & OPERATIONAL ACTION QUEUE

DISTRIBUTION BY DEPARTMENT (Department vs Total Bonus_Amount)

ACTIVITY TREND OVER HIRE_DATE (Month vs Total Bonus_Amount)

METHODOLOGY, AUDIT TRAIL & METRIC DICTIONARY

METHODOLOGICAL POLICIES & AGGREGATION AUDIT TRAIL

Policy #1: All analytical aggregations are computed locally in-browser using full-precision raw IEEE 754 float values.

Policy #2: No intermediate rounded values are ever re-aggregated across groups, metrics, or export layers.

Policy #3: Non-additive metrics (ratings, GPA, percentages, unit prices) are aggregated via weighted or arithmetic mean/median only; summation is strictly forbidden.

Policy #4: Semi-additive metrics (stock on hand, account balances) are point-in-time state indicators and must not be aggregated over time dimensions.

Policy #5: Missing, null, or empty string values are excluded on a per-metric basis without corrupting adjacent columns.

Policy #6: Zero AI hallucination guarantee: 100% deterministic code compilation with verified three-way engine parity.

CANONICAL METRIC DICTIONARY & ADDITIVITY TIERS

METRIC LABEL	FORMULA SPECIFICATION	ADDITIVITY	EXCLUSIONS	CANONICAL VALUE
Total Records Volume	`COUNTROWS`	**ADDITIVE**	0 omitted	**500**
Unique Entities (Emp_ID)	`COUNTDISTINCT(Emp_ID)`	**ADDITIVE**	0 omitted	**500**
Total Bonus_Amount	`SUM(Bonus_Amount)`	**ADDITIVE**	0 omitted	**5,059,110**
Mean Bonus_Amount	`AVERAGE(Bonus_Amount)`	**NON_ADDITIVE**	0 omitted	**10,118.22**
Median Bonus_Amount	`MEDIAN(Bonus_Amount)`	**ADDITIVE**	0 omitted	**10,118**
Total Annual_Salary	`SUM(Annual_Salary)`	**ADDITIVE**	0 omitted	**42,159,250**
Mean Annual_Salary	`AVERAGE(Annual_Salary)`	**NON_ADDITIVE**	0 omitted	**84,318.50**
Median Annual_Salary	`MEDIAN(Annual_Salary)`	**ADDITIVE**	0 omitted	**84,318.50**

INDEPENDENT MATHEMATICAL RECONCILIATION & REPRODUCIBILITY GUARANTEE

Source Hash: bab4e166 | Report Hash: e4c913f7 | Audit Timestamp: 2026-10-23T16:31:01.003Z

All metrics verified against independent Python float64 recalculation. Full raw precision preserved across all export targets.